

Project title: Retail Sales & Profitability Analysis Dashboard

Project subtitle: An end-to-end Excel data analytics project focused on sales performance, customer behavior, product profitability, discount analysis, and business recommendations.

Project Description: Analyzed 10,194 retail transactions using Microsoft Excel to evaluate sales, profitability, customer performance, product-level risk, discount patterns, and seasonal trends. Built an interactive dashboard with KPI cards, 16 analytical charts, slicers, business insights, and data-driven recommendations to support business decision-making.

Project Metadata	Field	Value
	Project Type	Data Analytics / Business Intelligence
	Tool	Microsoft Excel
	Dataset	Retail Superstore Sales
	Records Analyzed	10,194
	Primary Focus	Sales & Profitability Analysis
	Dashboard	Interactive Excel Dashboard

Business Problem: Retail businesses generate large volumes of transaction data, but raw sales records alone do not clearly reveal where revenue is being generated, which products and customers are profitable, where losses occur, or how discounts and seasonal patterns affect performance. The objective of this project was to transform raw retail transaction data into actionable business insights using Excel.

Project Objectives: Measure overall sales, profit, quantity, and profit margin.
Analyze sales and profitability across categories and sub-categories.
Evaluate customer sales, profitability, margins, and order frequency.
Identify high-performing and loss-making products.
Analyze the relationship between discount levels and profitability.
Identify sales trends, seasonal patterns, and areas requiring business attention.

Methodology

The project followed an end-to-end Excel analytics workflow: raw data inspection, data quality validation, data cleaning, exploratory analysis, KPI development, PivotTable analysis, customer and product analysis, discount analysis, time-based analysis, dashboard development, business insights, and recommendations.

Tools & Technologies

Field	Value
Tool	Microsoft Excel
Excel Features	Excel Tables
Analysis Tools	PivotTables & PivotCharts
Visualization	Excel Charts & Slicers
Data Preparation	Data Cleaning & Validation
Business Analysis	KPI Development & Business Analysis

Key Analysis Performed

Analysis Area	What Was Analyzed
Overall Performance	Total sales, profit, quantity, and profit margin
Category Analysis	Sales, profit, and margin by category
Sub-Category Analysis	Sales, profit, margin, and discount performance
Time Analysis	Annual, monthly, and quarterly sales trends
Customer Analysis	Customer sales, profit, margin, and order frequency
Product Analysis	Top-performing, high-sales, and loss-making products
Discount Analysis	Discount levels and their relationship with profitability
Risk Analysis	High-sales loss-making products and profitability risks
Dashboard Analysis	KPI cards, 16 analytical charts, and interactive slicers

Finding Area	Key Finding
Overall Performance	Total sales of ₹23,26,534.35 and total profit of ₹2,92,296.81 produced an overall profit margin of 12.56%.
Category Performance	Technology generated the highest sales and profit, with a 17.45% profit margin.

Key Findings

Furniture Profitability	Furniture generated ₹7,54,747.76 in sales but only ₹19,730.00 profit, resulting in a 2.61% margin.
Tables	Tables generated ₹2,08,020.18 in sales but recorded a loss of ₹17,753.21 and a -8.53% margin.
Sales Trend	Sales increased from ₹4,94,040.21 in 2023 to ₹7,45,567.53 in 2026, despite a decline in 2024.
Customer Profitability	High customer sales did not always correspond to profitability; Sean Miller generated ₹25,043.05 in sales but recorded a ₹1,980.74 loss.
Product Risk	Several high-sales products generated negative profit, requiring further investigation into pricing, discounts, and costs.
Discount Analysis	Higher discount levels were associated with lower profitability in this dataset, particularly at extreme discount levels.
Seasonal Pattern	Q4 generated the highest aggregated quarterly sales at approximately ₹8.90 lakh, while Q1 generated the lowest.

Recommendations

Recommendation Area	Recommended Action
Tables Profitability	Review pricing, discount levels, and cost structure for Tables.
Loss-Making Products	Investigate high-sales products generating negative profit.
Discount Strategy	Review discount policies, especially at higher discount levels.
High-Margin Products	Identify and support products and sub-categories with consistently strong margins.
Customer Profitability	Evaluate customers using profitability and margins, not sales volume alone.
Seasonal Planning	Align inventory and sales planning with stronger second-half sales patterns.

Furniture Performance	Investigate Furniture profitability while maintaining growth in Technology and Office Supplies.
Customer Relationships	Evaluate customer relationships using both order frequency and profitability.
Demand Planning	Monitor large month-to-month sales changes to improve demand planning.
Q4 Planning	Strengthen category-level planning for the high-performing Q4 period.

Project Outcome

This project transformed 10,194 raw retail transaction records into a structured Excel analytics solution. The analysis identified key sales and profitability patterns across categories, customers, products, discounts, and time periods. An interactive dashboard was developed with KPI cards, 16 analytical charts, and slicers to support business performance monitoring. The project also produced documented insights and actionable recommendations focused on improving profitability, managing discounts, investigating loss-making products, and supporting business planning.